

Allocated teaching periods	5
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Competency

- To study the probability in daily life and solve the problems using mathematical structures

Learning Outcomes

- To introduce probability scale and solve the simple problems related to probability

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define probability - To tell the definition of sample space - To recall the probabilities of certain and impossible events - To tell the probability scale
2.	Understanding (U)	- To find the probability of an event - To solve the simple problems on empirical probability

Required Teaching Materials/ Resources

Coin, Dice, cards, spinner etc

Pre-knowledge: Possibility of raining, having birth of son or daughter etc

Teaching Activities

- Take a coin and ask the following questions
 - How many faces are there?
 - Which face can be surely placed when the coin is tossed?
 - Can both faces be shown at once?
 - What is the possibility of getting head?
 - What is the possibility of getting tail?
- Similarly, take a die and ask the following questions
 - How many faces are there?
 - Which face can be surely placed when the coin is tossed?
 - Is it possible to fix all six faces at a roll?

- (iv) What is the possibility of getting 1?
 - (v) What is the possibility of getting each face separately?
 - (vi) What is the possibility of getting an odd number?
 - (vii) What is the possibility of getting a prime number?
3. Also, discuss on the following questions
- (i) What is the probability of raining today?
 - (ii) What is the probability of having daughter from a pregnant woman?
 - (iii) What is the probability of getting A⁺ grade by Ram in Mathematics?
4. Discuss on the following terminologies with proper examples
- (i) Probability
 - It measures the chances of happening or not happening the event
 - The numerical measurement of the degree of certainty of the occurrence of events

Example: When a coin is tossed, it is 50/50 chance that the head or tail occurs. So, the probability of occurrence of head is $\frac{1}{2}$ and that of tail is also $\frac{1}{2}$.
 - (ii) Random experiment
 - The experiment is an action by which an observation is made.
 - The random experiment is an experiment whose outcome cannot be predicted or determined in advance.
 - Examples: tossing a coin, rolling a die, drawing a card from a well-shuffled pack of 52 playing cards etc.
 - (iii) Sample space
 - The results of random experiments are called outcomes.
 - The set of all possible outcomes in a random experiment is the sample space.
 - Sample space is usually denoted by S.
 - Example: While flipping a coin, the possible outcomes are head (H) or tail (T).
 $\therefore S = \{H, T\}$
 - (iv) Event
 - Any non-empty subset of a sample space S is called an event.
 - Example: When a die is rolled, the sample space (S) = {1, 2, 3, 4, 5, 6}
 Here, {1}, {2}, {1, 2, 3}, {2, 4, 6} etc are some events.
 - 'S' itself is the sure event and empty subset ϕ is an impossible event.
 - (v) Exhaustive and favourable cases
 - The number of possible outcomes of a random experiment is the exhaustive cases.
 Example: When a coin is tossed twice, then $S = \{HH, HT, TH, TT\}$ and exhaustive cases = 4.
 - The number of desirable (expected) outcomes in a random experiment is the favourable cases. Example: When tossing a coin, the favourable cases of getting head = 1 and that of tail = 1
 - (vi) Equally likely events
 - Two or more events are said to equally likely events, if the chance of occurring any one event is equal to the chance of occurring other events.
 - Example: While tossing a coin, the change of occurring head and tail is equal.

(vii) Mutually exclusive events

- Two or more events in a sample space are called mutually exclusive if the occurrence of one event excludes the occurrence of other.
- Two events A and B of a sample space S are mutually exclusive if $A \cap B = \phi$.
- Example: While tossing a coin, the occurrence of head excludes the occurrence of tail or $H \cap T = \phi$

(viii) Dependent and independent events

- Two or more events are said to be dependent if the occurrence of one event affects the occurrence of the other events.
- Example: While drawing a marble in successive trials from a bag containing 3 green and 6 blue marbles without replacement, getting any one coloured ball in the first trial affects to draw another ball in the second trial.
- Two or more events are said to be independent if the occurrence of one event does not affect the occurrence of the other events.
- Example: While tossing a coin twice or more, the occurrence of any one event in the first toss does not affect the occurrence of any events in other trials.

5. Explain the probability of an event (E) in an exhaustive case/sample space (S) as

$$P(E) = \frac{\text{Favourable number of cases}}{\text{Exhaustive number of cases}} = \frac{n(E)}{n(S)}$$

$$\text{and } P'(E) = 1 - P(E)$$

6. Show the coin, die and pack of 52 playing cards, and make clear about the facts on coin, dice and playing cards.
7. Discuss about the probabilities of different events including certain and impossible events
8. With examples, make the students discover probability scale
9. Divide the students into groups and give 1/1 coin to each group. Tell them to flip the coin 20/20 times and record the outcomes.

Outcomes	Head(H)	Tail (T)
Frequency	...	

10. Then ask to find the probability of getting head and getting tail.
11. With above experiment, discuss about the empirical probability.

Solution of selected questions from Excel in Mathematics

1. In a class of 40 students, 3 boys and 5 girls wear spectacles. If a teacher called one of the students randomly in the office, find the probability that this student is wearing the spectacle.

Solution:

Here, total number of students, $n(S) = 40$

Favourable number of cases of wearing glasses, $n(E) = 3 + 5 = 8$

$$\text{Now, } P(E) = \frac{n(E)}{n(S)} = \frac{8}{40} = \frac{1}{5}$$

Hence, the probability that the student wearing spectacle is $\frac{1}{5}$.

2. **There are 40 students in a class with roll numbers from 1 to 40. The roll number of Bhurashi is 18. If a teacher calls one student with roll number exactly divisible by 3 to do a problem on blackboard, what is the probability that Bhurashi will be selected? Also, find the probability that she will not be selected.**

Solution:

Here,

Set of roll numbers which are exactly divisible by 3, $(S) = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39\} \therefore n(S) = 13$

Favourable number of cases, $n(E) = 1$

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{1}{13}$ Thus, the probability that Bhurashi can be selected is $\frac{1}{13}$.

Hence, the probability that Bhurashi will not be selected, $P'(E) = 1 - \frac{1}{13} = \frac{12}{13}$.

3. **A card is drawn at random from a deck of 52 cards, what is the probability that the card**

(i) **is an ace**

(ii) **is an ace of spade**

(iii) **is a black ace**

Solution:

Here, total number of cards, $n(S) = 52$

- (i) Favourable number of cases of getting ace, $n(E_1) = 4$

$$\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{4}{52} = \frac{1}{13}$$

Thus, the probability of getting an ace is $\frac{1}{13}$.

- (ii) Favourable number of cases of getting ace of spade, $n(E_2) = 1$

$$\therefore P(E_2) = \frac{n(E_2)}{n(S)} = \frac{1}{52}$$

Thus, the probability of getting an ace of spade is $\frac{1}{52}$.

- (iii) Favourable number of cases of getting black ace, $n(E_3) = 2$

$$\therefore P(E_3) = \frac{n(E_3)}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

Thus, the probability of getting a black ace is $\frac{1}{26}$.

4. **A card is drawn at random from a well shuffled pack of 52 cards. Find the probability that the card will be black or face cards.**

Solution:

Here, total number of cards, $n(S) = 52$

Favourable number of cases of getting black cards = 26

Favourable number of cases of getting faced cards = 12

Favourable number of cases of getting black faced cards = 6

Total number of favourable cases, $n(E) = 26 + 12 - 6 = 32$

$$\text{Now, } P(E) = \frac{n(E)}{n(S)} = \frac{32}{52} = \frac{8}{13}$$

Thus, the probability that the card will be black or faced card is $\frac{8}{13}$.

5. One card is drawn at random from the number cards numbered from 10 to 21. Find the probability that the card may be prime or even numbered card.

Solution:

Here, Sample space (S) = {10, 11, ..., 21}

$$\therefore n(S) = (21 - 10) + 1 = 12$$

Favourable number of cases of prime numbers = 4

Favourable number of cases of even numbers = 6

Total number of favourable cases, $n(E) = 4 + 6 = 10$

$$\text{Now, } P(E) = \frac{n(E)}{n(S)} = \frac{10}{12} = \frac{5}{6}$$

Hence, the probability that the card may be prime or even numbered card is $\frac{5}{6}$.

6. Glass tumblers are packed in cartons, each containing 12 tumblers. 200 cartons were examined for broken glasses and the results are given in the table below:

No. of broken glasses	0	1	2	3	4	More than 4
Frequency	164	20	9	4	2	1

If one carton is selected at random, what is the probability that:

- it has no broken glass?
- it has broken glasses less than 3?
- it has broken glasses more than 1?
- it has broken glasses more than 1 and less than 4?

Solution:

Here, total number of cartons, $n(S) = 200$

- (i) The number of cartons that have no broken glass = $n(E_1) = 164$

$$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{164}{200} = \frac{41}{50} = 0.82$$

- (ii) The number of cartons that have less than 3 broken glasses

$$n(E_2) = 164 + 20 + 9 = 193$$

$$\therefore P(E_2) = \frac{n(E_2)}{n(S)} = \frac{193}{200} = 0.965$$

- (iii) The number of cartons that have more than 1 broken glasses

$$n(E_3) = 9 + 4 + 2 + 1 = 16$$

$$\therefore P(E_3) = \frac{n(E_3)}{n(S)} = \frac{16}{200} = 0.08$$

- (iv) The number of cartons that have more than 1 and less than 4 broken glasses, $n(E_4) = 9 + 4 = 13$

$$P(E_4) = \frac{n(E_4)}{n(S)} = \frac{13}{200} = 0.065$$

7. From a pack of playing cards, two cards are taken, which are not hearts. They are not replaced, and the remaining cards are shuffled. What is the probability that the next card drawn is heart?

Solution:

Here, total number of cards, $n(S) = 52 - 2 = 50$

Favourable number of cases of getting the cards of hearts, $n(E) = 4$

$$\text{Now, } P(E) = \frac{n(E)}{n(S)} = \frac{4}{50} = \frac{2}{25}$$

8. **Three athletes A, B and C are to run a race. B and C have equal chances of winning, but A is twice as likely to win as either. Find the probability of each athlete winning.**

Solution:

Here, let the probability of winning the race by B be x

According to question, probability of winning the race by C = x and the probability of winning the race by A = $2x$

Now, $P(A) + P(B) + P(C) = 1$ or, $2x + x + x = 1$ or, $4x = 1$ $\therefore x = \frac{1}{4}$
Hence, the probability of winning the race by A = $2x = 2 \times \frac{1}{4} = \frac{1}{2}$ and the probability of winning the race by B and C is $\frac{1}{4}$ each.

9. **A man has 3 pairs of black socks and 2 pair of brown socks. If he dresses hurriedly in the dark, find the probability that**
- the first sock he puts on is brown.**
 - the first sock he puts on is black.**
 - after he put on a black sock, he will then put on another black sock.**
 - that after he has first put on a brown sock, the next sock will also be brown.**

Solution:

Here, total number of socks, $n(S) = 2(3 + 2) = 10$

(i) The number of brown socks = $n(E_1) = 4$ $\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{4}{10} = \frac{2}{5}$

(ii) The number of black socks = $n(E_2) = 6$ $\therefore P(E_2) = \frac{n(E_2)}{n(S)} = \frac{6}{10} = \frac{3}{5}$

(iii) After putting on a black sock, the number of black socks for the next trial = $n(E_3) = 5$
and total number of socks, $n(S) = 10 - 1 = 9$

$$\therefore P(E_3) = \frac{n(E_3)}{n(S)} = \frac{4}{50} = \frac{5}{9}$$

(iv) After putting on a brown sock, the number of brown socks for the next trial = $n(E_4) = 3$ and total number of socks, $n(S) = 10 - 1 = 9$

$$\therefore P(E_4) = \frac{n(E_4)}{n(S)} = \frac{3}{9} = \frac{1}{3}$$

Extra Questions

- Define sample space. What is the probability of a certain event?
- A die is rolled once. What is the probability that the digit turn off is a prime number.
[Ans: $\frac{1}{2}$]
- A bag contains a dozen of identical balls. Among them, 3 are red, 5 are green and the rest are white. If a ball is randomly drawn, what is the probability of getting the white ball?
[Ans: $\frac{1}{3}$]
- A card is drawn at random from a well shuffled pack of 52 cards. Find the probability that the card will be black king or red queen.
[Ans: $\frac{1}{13}$]
- Three athletes A, B and C are to run a race. If A is twice as likely to win B and B is thrice as likely to win C. Find the probability of each athlete winning.
[Ans: $\frac{3}{5}, \frac{3}{10}, \frac{1}{10}$]